

PATIENT SCREENING REPORT

Retinal Edge Triage — Explainable AI-Assisted Diabetic Retinopathy Screening for Rural Healthcare
Screening Date & Time: 11-Sep-2026 19:01 UTC | Reference ID: SCR-2026-000187

Patient ID: PAT-DR-0190	Full Name: Ramesh Gowda	Age / Gender: 58 / Male
Screening ID: SCR-2026-000187	Location: Mysuru Rural PHC	Diabetes Type / Duration: Type 2 (8 years)

1. Retinal Image Acquisition & Quality Gate

Eye Evaluated: Right Eye	Image Source: Portable Fundus Camera	Quality Score: 65.6%	Quality Status: BORDERLINE
Focus / Blur: 32.3%	Illumination: 46.3%	Vessel Visibility: 74.8%	Field of View: 91.5%

2. Diabetic Retinopathy Screening & Detection Result

Screened Eye:	Right Eye	DR Grade & Severity:	Grade: 3 — Severe NPDR
Model Confidence:	30.4%	DME Risk Status:	Possible DME detected
Class Probabilities:	DR 0: 5.1% DR 1: 19.9% DR 2: 27.2% DR 3: 30.4% DR 4: 17.4%	DR Status:	Diabetic Retinopathy: DETECTED
Result Explanation:	Severe non-proliferative diabetic retinopathy with multi-quadrant hemorrhages (9) and vascular beading.	Recommended Action:	Severe diabetic retinopathy requires urgent ophthalmic evaluation within 1 month.

3. Previous vs Current Screening Comparison

Previous Screening:	Date: 2026-09-11 Grade: DR 2	Current Screening:	Date: 11-Sep-2026 19:01 UTC Grade: DR 3
Overall Trend:	WORSENE	Recommendation:	Specialist review recommended due to disease progression.

4. Human-in-the-Loop Ophthalmologist Adjudication

Doctor Decision:	ACCEPTED	Reviewing Doctor:	Dr. Rajesh Varma, MS (Ophthal)
Final Grade Assigned:	Grade 3	Override Justification:	None
Clinical Notes:	Severe NPDR confirmed with multiple blot hemorrhages. Expedited laser/anti-VEGF consultation mandated.	HCW Instructions:	None

5. Risk Triage & Clinical Disposition

Risk Category:	HIGH RISK	Referral Priority:	URGENT
Referral Reference ID:	REF-2026-00124	Follow-up Schedule:	3 Months
Clinical Justification:	Moderate DR with DME risk identified.	Referral Status:	PENDING

6. Recommended Eye-Care Facility (Referral Destination)

Hospital / Clinic Name:	ABC Eye Hospital	Distance:	2.4 km away
Hospital Address:	Sayyaji Rao Road, Medar Block, Yadavagiri, Mysuru, Karnataka 570020	Contact Number:	+91 821 241 9000
Reason for Referral:	Severe NPDR requires ophthalmic evaluation based on AI screening result.	Navigation / Directions:	View Map & Directions

7. Patient Care & Lifestyle Guidance

Guidance: Severe diabetic retinal changes were detected. Specialist eye evaluation is recommended.

- Take prescribed diabetes medicines regularly.
- Monitor blood glucose as advised by your healthcare provider.
- Follow a balanced diet rich in vegetables and whole grains.
- Stay physically active as advised by your healthcare provider.
- Avoid smoking and tobacco products.
- Attend recommended comprehensive eye check-ups.
- Do not stop prescribed medicines without consulting your doctor.

NOTICE: Patient care and lifestyle guidance support general wellness and glycemic control, but do not cure or reverse diabetic retinopathy (DR 1–4) and cannot replace examination by an eye specialist.

STATUTORY CLINICAL NOTICE: This referral recommendation is based on the screening assessment. Final clinical management should be determined by a qualified healthcare professional. AI predictions DO NOT constitute a definitive medical diagnosis. Final clinical interpretation and management must always be performed by a qualified ophthalmologist.